Practical - 9

Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

A . Theory

Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.

1. **Is password_hash() used before storing passwords?**
   Yes, `password_hash()` is used before storing the password in the database. This helps to store the password securely instead of storing it in plain text.
2. **Is duplicate email/username checked?**
   Yes, duplicate username and email are checked before inserting the new user. If the username or email already exists, registration is not allowed.
3. **Are frontend and backend validations consistent?**
   Yes, both frontend and backend validations are used. Frontend validation checks the input before submitting the form, while backend validation checks the data again before storing it in the database.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 9.1 : Registration Form

Registration successful!

StudentHub Registration

Username: Dhyani

Email: abc@gmail.com

Password: ••••••••

Confirm Password: ••••••••

Register

Figure 9.2 : Database Students

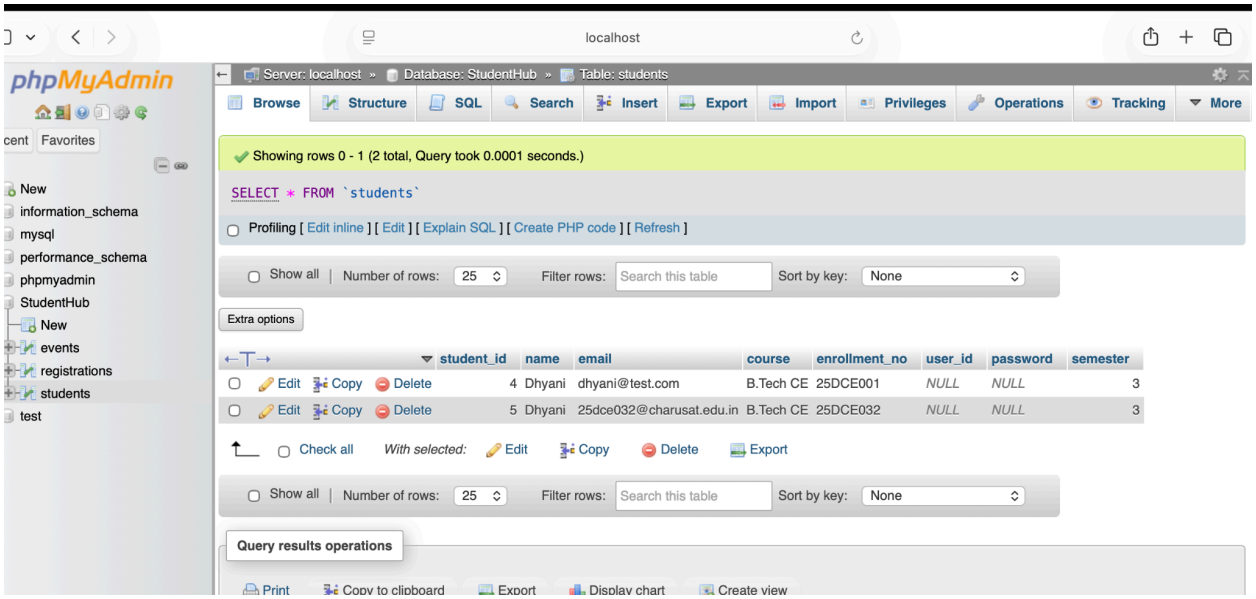


Figure 9.3 : Database users

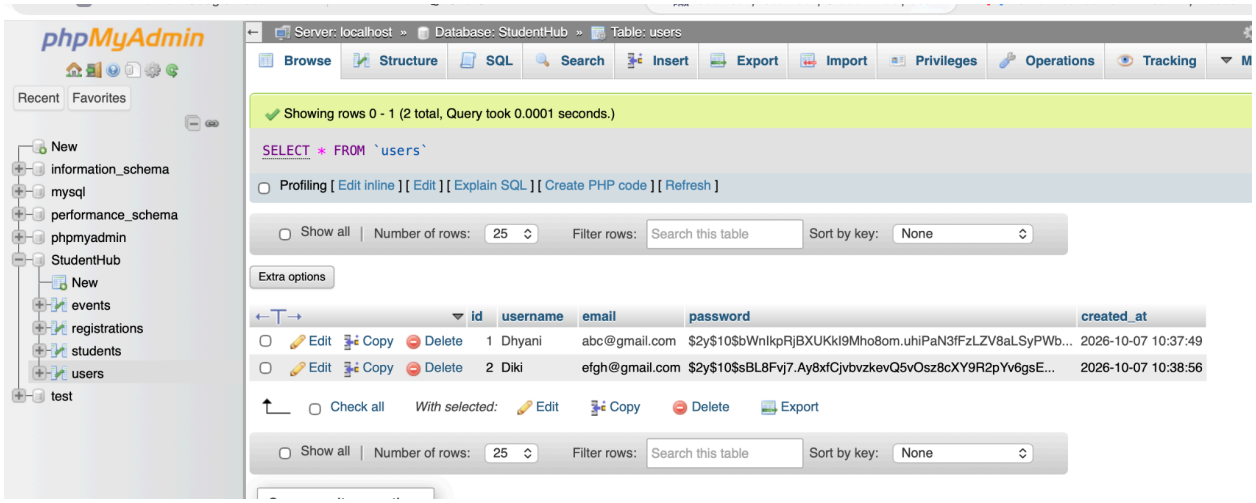
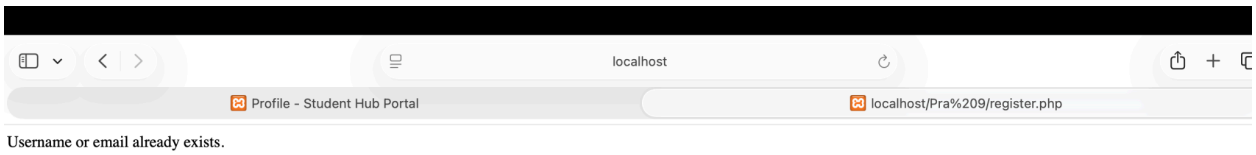


Figure 8.4 : Veryfing email and username



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

During this practical, I faced some difficulties in connecting the registration form with the MySQL database and storing user details properly. I also had to handle duplicate email addresses and usernames so that the same details could not be registered again. Understanding how to use `password_hash()` to store passwords securely was another

challenge. After testing the form with different inputs, I was able to fix the issues and make the registration process work correctly.

F. Learning Outcomes:

Through this practical, I learned how to create a secure user registration system using PHP and MySQLi. I understood how to validate user input on both the frontend and backend, check for duplicate email addresses and usernames, and store user details in a database. I also learned the importance of password hashing for protecting users' passwords and displaying appropriate messages for successful registration or errors.

Faculty Remarks: